

Title: Requirements Table

Subtitle: Problem Statement #44 — Database Query Performance Profiler

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Subtitle 2: Lab 1: Requirements Engineering & UML Use-Case Modelling

Stakeholders / Actors: Database Administrator, Backend Lead.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse Postgres/MySQL EXPLAIN plans, highlight sequential table scans on large tables, and recommend optimal index column definitions.	High	Pass: Missing composite index identified on filtered columns. Fail: Query plan with sequential scan marked optimized.	Core value proposition of the tool — turns raw execution plans into actionable index recommendations.
FR-002	Functional	The system shall ingest slow query logs from Postgres and MySQL sources on a continuous or scheduled basis and store each entry with timestamp, query text, and execution duration.	High	Pass: A log line written to the monitored source appears as a stored record with correct timestamp and duration within 5 minutes. Fail: Entry is missing or duration differs by more than 1%.	Ingestion is the entry point of the pipeline; without reliable capture, no downstream analysis is possible.
FR-003	Functional	The system shall generate a weekly performance optimization digest summarizing top slow queries, recommended indexes, and estimated performance impact, and deliver it to	High	Pass: A digest is generated every 7 days containing at least the top 10 slow queries with recommendations and is received by the DBA and Backend Lead. Fail: Digest missing, late by more than 24 hours, or missing recommendation data.	Converts raw analysis into a recurring, actionable artifact that drives optimization work without manual effort.

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		subscribed stakeholders.			
FR-004	Functional	The system shall provide a dashboard that displays the top N slowest queries over a selectable time range, filterable by database instance and query pattern.	Medium	Pass: Selecting a time range and filter updates the displayed query list within 3 seconds and matches the underlying stored data. Fail: Dashboard shows stale or unfiltered data after selection.	Gives the Backend Lead and DBA on-demand visibility between weekly digests, supporting faster investigation.
FR-005	Functional	The system shall allow a Database Administrator to configure the slow-query threshold (in milliseconds) used to classify a query as "slow" for a given database instance.	Medium	Pass: After saving a new threshold value, subsequent ingested queries are classified using the new value and the change is reflected in the next digest. Fail: Old threshold continues to be applied after save.	Different workloads have different performance baselines; a fixed threshold would misclassify queries across environments.
NFR-001	Nonfunctional	Slow query log ingestion must handle up to 5,000 log records per minute without dropping entries or slowing the host database.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Any entry dropped or host DB latency increases beyond baseline during test.	Performance & Security — the profiler must not become a bottleneck or risk for the production databases it monitors.
NFR-002	Nonfunctional	All ingested query logs and execution plans, which may contain sensitive query parameters, shall be encrypted at rest (AES-256) and access shall be restricted by role (Database Administrator, Backend Lead).	High	Pass: Stored log files are unreadable without the decryption key, and a user without an assigned role receives an access-denied response when querying the API/dashboard. Fail: Data readable at rest without key, or unauthorized role can view logs.	Security — query logs can expose table structure, filter values, and business logic; access must be controlled.

